GNSS Principles Experiment Report

Week 7, Lab 6, Application of Satellite Signal Simulator , April.01

1 Write the basic working principle of a satellite signal simulator. (2)Describe the main error model used in a satellite signal simulator.

A satellite signal simulator is a laboratory tool designed to generate and transmit synthetic satellite signals, allowing controlled testing of navigation and communication equipment. Its basic working principle can be summarized as follows:

(a) GNSS simulator

- **Signal Generation:** The simulator recreates the RF signals that satellites transmit by replicating the specific modulation schemes (e.g., Binary Phase-Shift Keying (BPSK) for GPS) and spreading codes (e.g., GPS C/A-code). It digitally generates the baseband waveform corresponding to each satellite.
- **Scenario Modeling:** Using pre-programmed satellite orbit data and receiver trajectory information, the simulator constructs a complete *scenario* that mimics the real-world environment. This includes generating multiple satellite signals simultaneously to simulate a constellation and incorporating factors such as satellite geometry and dynamic motion.
- **Error Injection:** To achieve realism, the simulator injects a variety of errors into the generated signals. These include:
  - **Satellite Clock and Ephemeris Errors:** Systematic biases in satellite timing and orbital parameters.
  - **Atmospheric Delays:** Models for ionospheric and tropospheric delays that affect the signal's propagation speed.

  - **Multipath Effects:** Delayed and attenuated copies of the signal are added to emulate reflections from surrounding objects.

- **RF Conversion and Output:** The digitally generated and error-modified signals are then converted to analog form, upconverted to the appropriate RF band (e.g., L-band for GPS), and finally transmitted through an antenna. This allows the signals to be received by actual satellite receiver hardware.

2 Using a variable attenuator and a satellite positioning receiver, test and record the changes in the carrier-to-noise ratio (C/N_0) of the satellite signals observed by the receiver under different attenuation levels.

In this experiment, a variable RF attenuator was used to control the power of the GNSS signal being received, in order to observe the change in the Carrier-to-Noise density ratio (C/N_0) as detected by a satellite receiver.

Experiment Setup

- A GNSS signal receiver connected to a simulator.
- A variable RF attenuator capable of applying attenuation in discrete steps: 1, 2, 4, 8, 16, and 20 dB.
- Monitoring software to visualize the satellite signal strength (C/N_0).

Procedure and Observations

Case 1: No Attenuation Applied
Initially, no buttons on the attenuator were pressed. This means no attenuation was introduced. As shown in the figure below, satellite signals such as G21 maintained high C/N_0 values:

$$C/N_0 \approx 51.0 \text{ dB-Hz}$$

Case 2: 31 dB Attenuation Applied
All attenuation buttons were pressed except for the 20 dB button, resulting in a total attenuation of:

$$1 + 2 + 4 + 8 + 16 = 31 \text{ dB}$$

This significantly reduced the signal strength received. Satellite G18, for example, dropped to:

$$C/N_0 \approx 36.0 \text{ dB-Hz}$$

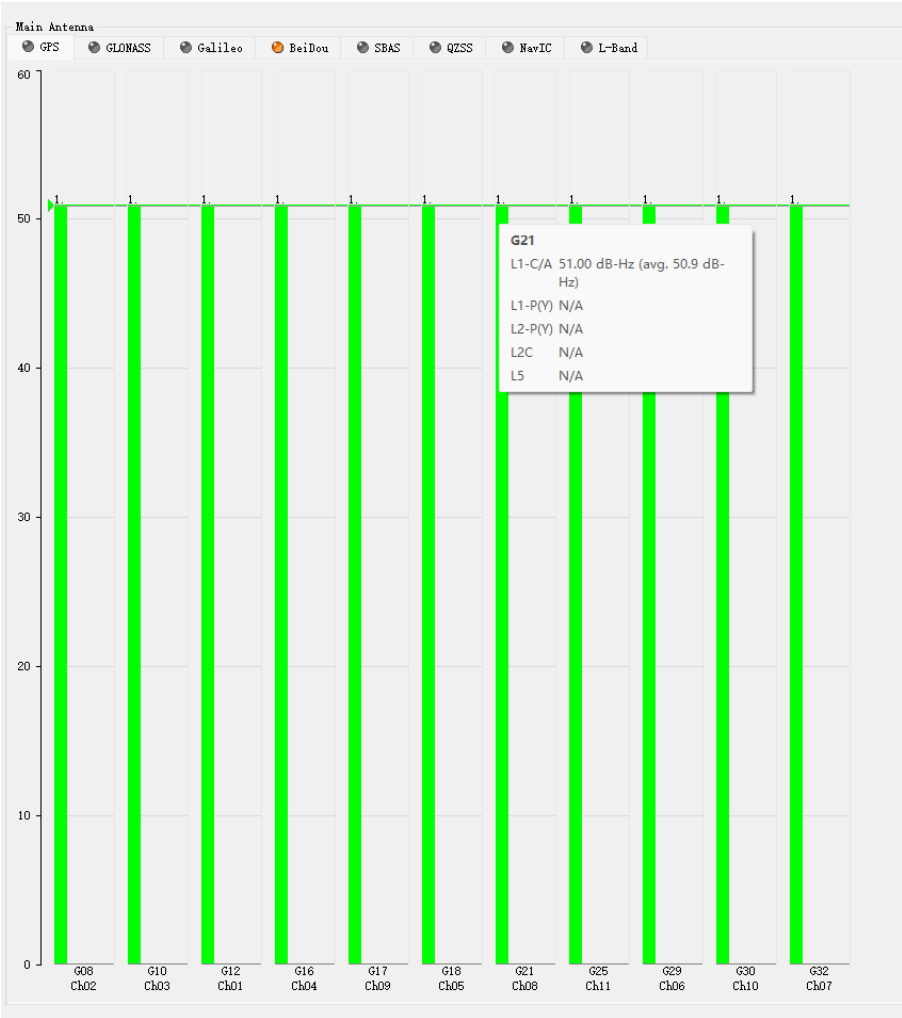


Figure 2: Satellite signals with no attenuation (G21 C/N_0 = 51.0 dB-Hz)

Total Attenuation (dB)	Example Satellite	C/N_0 (dB-Hz)
0 (No attenuation)	G21	51.0
31 dB	G18	36.0

Table 1: Effect of RF attenuation on satellite signal strength

Results Summary

Conclusion

This experiment demonstrates a direct relationship between RF signal attenuation and reduced signal strength. As the attenuation increased, the C/N_0 detected by the receiver decreased significantly. This mimics conditions such as signal loss due to obstruction, interference, or distance from the satellite, making it a valuable test setup for robustness analysis of GNSS systems.

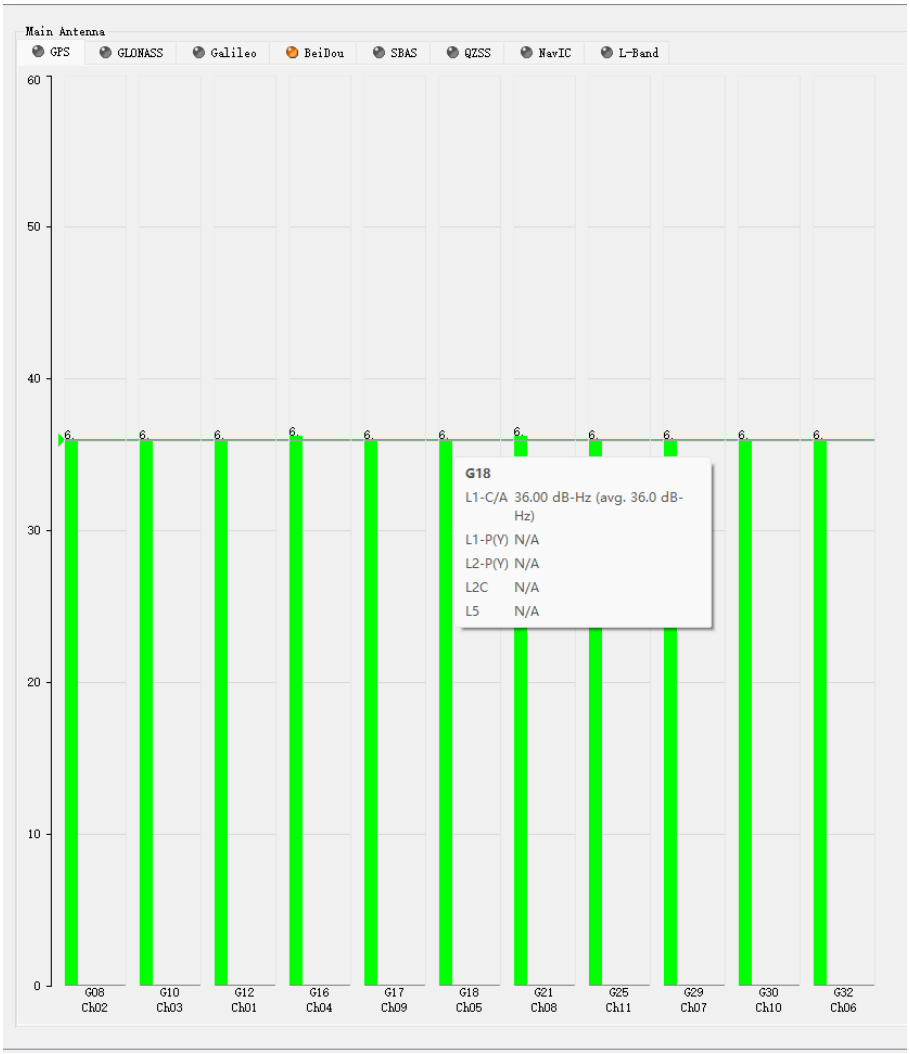


Figure 3: Satellite signals under 31 dB attenuation (G18 C/N_0 = 36.0 dB-Hz)

2.1 Capture (Acquisition)

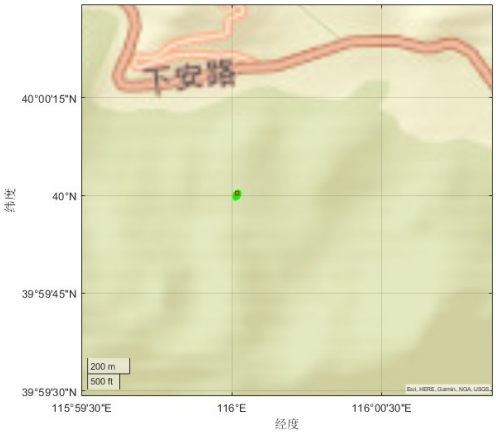
The first step involves acquiring satellite signals by identifying their Pseudo-Random Noise (PRN) codes. The acquisition process searches for visible satellites and measures their signal strength.

2.2 Tracking

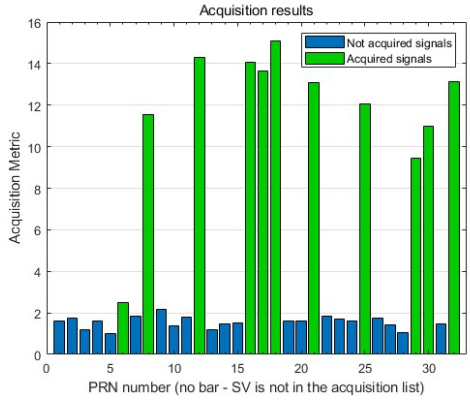
Once the signals are acquired, the receiver tracks the carrier phase and code phase using Phase-Locked Loops (PLL) and Delay-Locked Loops (DLL). The tracking process extracts navigation message bits and maintains signal lock.

2.3 Positioning

The final step computes the receiver's position using the tracked signals. The position is estimated in the Universal Transverse Mercator (UTM) coordinate system, and satellite geometry is evaluated using a sky plot.



(a) Map showing the receiver's location near 40°N, 116°E.



(b) Histogram of acquisition results showing acquired signals (green) vs. total signals (blue) for different PRNs.

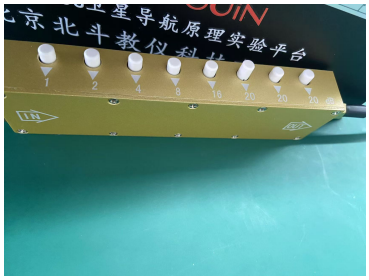
Figure 4: Acquisition phase results.

3 Results and Discussion

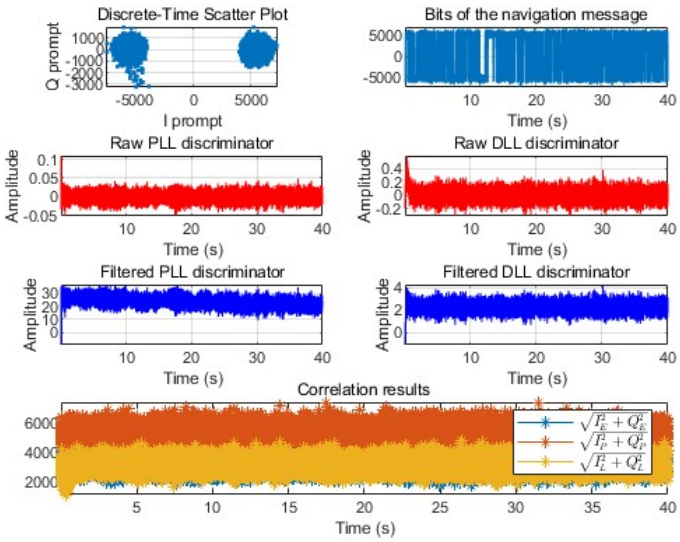
- **Capture**: The receiver successfully acquired signals from multiple satellites, as shown in Figure 4b. Most PRNs were detected, with acquisition metrics peaking around 14–15.
- **Tracking**: The tracking phase maintained signal lock, as evidenced by the scatter plot and correlation peaks in Figure 5b. The navigation message bits were successfully extracted.
- **Positioning**: The receiver's mean position was calculated at latitude 40°0.033035°N, longitude 116°0.01499°E, and height 123.6 m (Figure 6). Position variations were within ±10 m horizontally and ±20 m vertically. The mean PDOP of 3.6839 indicates acceptable satellite geometry.

4 Conclusion

The simulated GNSS signal was successfully processed through capture, tracking, and positioning. The receiver accurately estimated its position near Beijing, China, with reasonable precision and satellite geometry. Future work could involve real-world signal testing to validate these results.



(a) Signal attenuator device used for simulation.



(b) Tracking results: scatter plot, navigation message bits, raw and filtered PLL/DLL discriminators, and correlation results.

Figure 5: Tracking phase results.

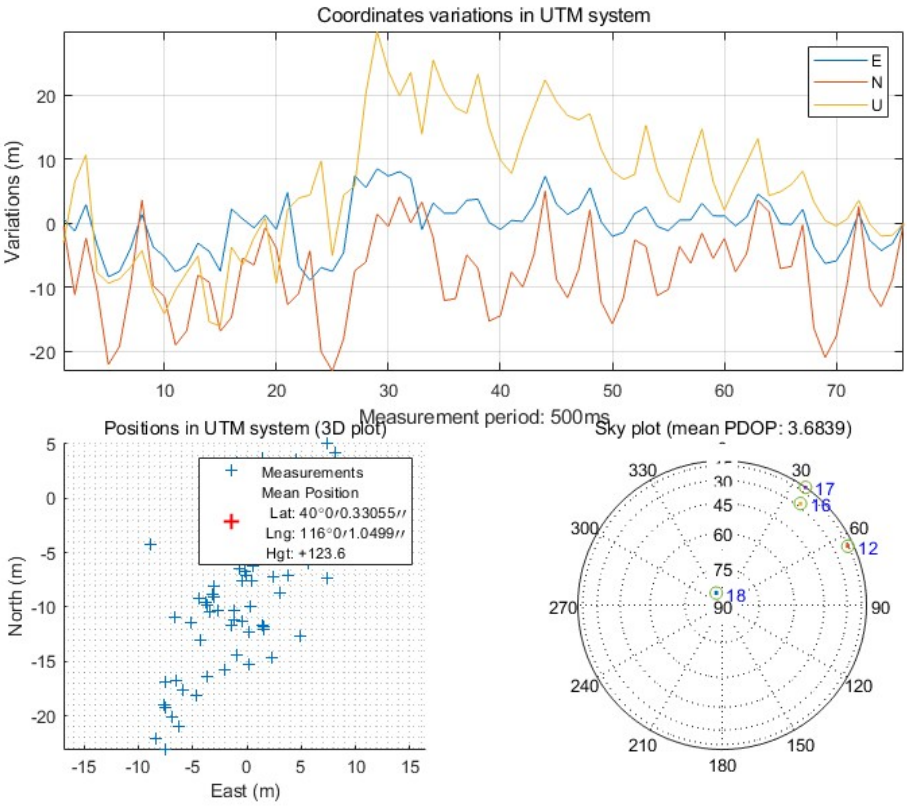


Figure 6: Positioning results: coordinate variations in UTM system, 3D scatter plot of positions (mean position at latitude 40°0.033035°N, longitude 116°0.01499°E, height 123.6 m), and sky plot with mean PDOP of 3.6839.